

# **Mathematics**

## **A Cultural Heritage**

### **Lecture 1**

#### **Introduction: Numbers**

# Course Topics: Numbers

## Numbers



| Bank Account Information |            |       |
|--------------------------|------------|-------|
| Starting Balance         | 01/12/2014 | 23000 |
| Deposit                  | 07/12/2014 | 1100  |
| Withdrawal               | 13/12/2014 | 1300  |
| New Balance              | 01/01/2015 | 22800 |

What about “-1300”?

What does “3” mean?



# Course Topics: Numbers

## Number Systems of different cultures

|                       | 1      | 2     | 3      | 4       | 5        | 6         | 7          | 8           | 9            |
|-----------------------|--------|-------|--------|---------|----------|-----------|------------|-------------|--------------|
| UNITS                 | 𐍌      | 𐍌𐍌    | 𐍌𐍌𐍌    | 𐍌𐍌𐍌𐍌    | 𐍌𐍌𐍌𐍌𐍌    | 𐍌𐍌𐍌𐍌𐍌𐍌    | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌    | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌    | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌    |
| TENS                  | 𐍌𐍌     | 𐍌𐍌𐍌   | 𐍌𐍌𐍌𐍌   | 𐍌𐍌𐍌𐍌𐍌   | 𐍌𐍌𐍌𐍌𐍌𐍌   | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌   | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌   | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌   | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌   |
| HUNDREDS              | 𐍌𐍌𐍌    | 𐍌𐍌𐍌𐍌  | 𐍌𐍌𐍌𐍌𐍌  | 𐍌𐍌𐍌𐍌𐍌𐍌  | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌  | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌  | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌  | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌  | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌  |
| THOUSANDS             | 𐍌𐍌𐍌𐍌   | 𐍌𐍌𐍌𐍌𐍌 | 𐍌𐍌𐍌𐍌𐍌𐍌 | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌 | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌 | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌 | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌 | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌 | 𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌𐍌 |
| TENS OF THOUSANDS     | 𐍌𐍌𐍌𐍌𐍌  |       |        |         |          |           |            |             |              |
| HUNDREDS OF THOUSANDS | 𐍌𐍌𐍌𐍌𐍌𐍌 |       |        |         |          |           |            |             |              |

*Egyptian hieratic numerals of the New Kingdom (1600-1000BC) used more symbols than before, making numbers more compact but harder to learn to use.*

|    |       |    |         |    |        |    |         |     |          |
|----|-------|----|---------|----|--------|----|---------|-----|----------|
| 1  | I     | 21 | XXI     | 41 | XLI    | 61 | LXI     | 81  | LXXXI    |
| 2  | II    | 22 | XXII    | 42 | XLII   | 62 | LXII    | 82  | LXXXII   |
| 3  | III   | 23 | XXIII   | 43 | XLIII  | 63 | LXIII   | 83  | LXXXIII  |
| 4  | IV    | 24 | XXIV    | 44 | XLIV   | 64 | LXIV    | 84  | LXXXIV   |
| 5  | V     | 25 | XXV     | 45 | XLV    | 65 | LXV     | 85  | LXXXV    |
| 6  | VI    | 26 | XXVI    | 46 | XLVI   | 66 | LXVI    | 86  | LXXXVI   |
| 7  | VII   | 27 | XXVII   | 47 | XLVII  | 67 | LXVII   | 87  | LXXXVII  |
| 8  | VIII  | 28 | XXVIII  | 48 | XLVIII | 68 | LXVIII  | 88  | LXXXVIII |
| 9  | IX    | 29 | XXIX    | 49 | XLIX   | 69 | LXIX    | 89  | LXXXIX   |
| 10 | X     | 30 | XXX     | 50 | L      | 70 | LXX     | 90  | XC       |
| 11 | XI    | 31 | XXXI    | 51 | LI     | 71 | LXXI    | 91  | XCI      |
| 12 | XII   | 32 | XXXII   | 52 | LII    | 72 | LXXII   | 92  | XCH      |
| 13 | XIII  | 33 | XXXIII  | 53 | LIII   | 73 | LXXIII  | 93  | XCHH     |
| 14 | XIV   | 34 | XXXIV   | 54 | LIV    | 74 | LXXIV   | 94  | XCIV     |
| 15 | XV    | 35 | XXXV    | 55 | LV     | 75 | LXXV    | 95  | XC V     |
| 16 | XVI   | 36 | XXXVI   | 56 | LVI    | 76 | LXXVI   | 96  | XCVI     |
| 17 | XVII  | 37 | XXXVII  | 57 | LVII   | 77 | LXXVII  | 97  | XC VII   |
| 18 | XVIII | 38 | XXXVIII | 58 | LVIII  | 78 | LXXVIII | 98  | XCVIII   |
| 19 | XIX   | 39 | XXXIX   | 59 | LIX    | 79 | LXXIX   | 99  | XCIX     |
| 20 | XX    | 40 | XL      | 60 | LX     | 80 | LXXX    | 100 | C        |

200 CC 300 CCC 400 CD 500 D 600 DC 700 DCC 800 DCCC 900 CM 1000 M

|   |   |   |                 |     |    |
|---|---|---|-----------------|-----|----|
| 0 | 〇 | 零 |                 |     |    |
| 1 | 一 | 壹 | 10              | 十   | 拾  |
| 2 | 二 | 貳 | 20              | 廿/廿 | 貳拾 |
| 3 | 三 | 叁 | 30              | 卅   | 參拾 |
| 4 | 四 | 肆 | 40              | 卌   | 肆拾 |
| 5 | 五 | 伍 |                 |     |    |
| 6 | 六 | 陸 | 10 <sup>2</sup> | 百   | 佰  |
| 7 | 七 | 柒 | 10 <sup>3</sup> | 千   | 仟  |
| 8 | 八 | 捌 | 10 <sup>4</sup> | 万   | 萬  |
| 9 | 九 | 玖 | 10 <sup>8</sup> | 亿   | 億  |

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| Possible Progression |   |   |   |   |   |   |   |   |   |  |
|----------------------|---|---|---|---|---|---|---|---|---|--|
| 0                    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |  |
| 0                    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |  |
| 0                    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |  |

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# Course Topics: Numbers

## Different kinds of numbers

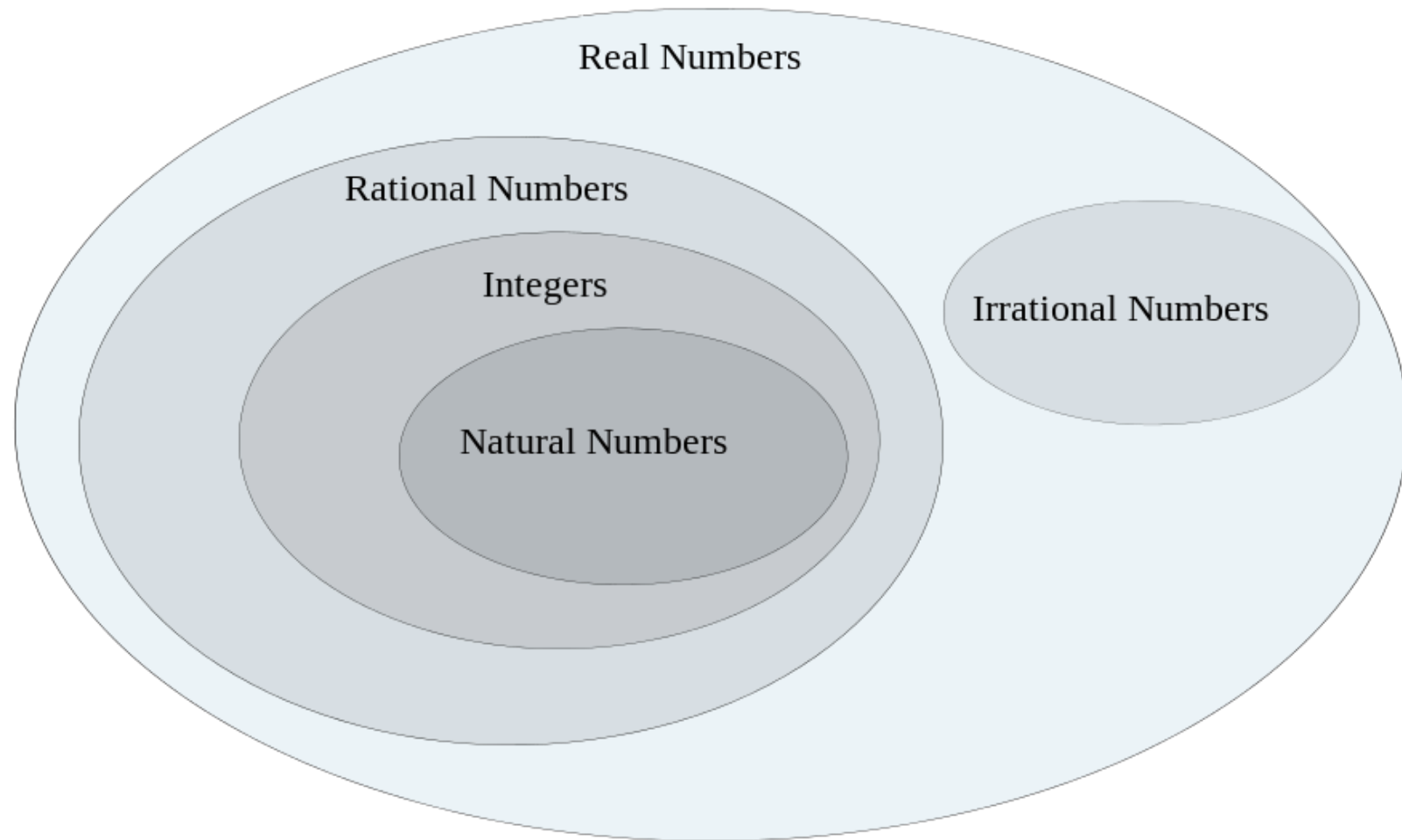
- Whole/natural numbers: **1, 2, 3, ...**
- Fractions/rational numbers:  $\frac{1}{2}, \frac{4}{9}, \frac{11}{13}, \dots$
- Irrational numbers:  $\sqrt{7}, \pi \approx 3.14159 \dots$
- Negative numbers: **-3, -7, -10, ...**
- The number **0**
- Imaginary numbers:  $i = \sqrt{-1}$



# Course Topics: Numbers

- Complex Numbers:  $3 + 2\sqrt{-1}, 7 - i$
- Matrices:  $\begin{bmatrix} 3 & 7 \\ 4 & 2 \end{bmatrix}, \begin{bmatrix} \frac{5}{3} & \frac{2}{7} \\ 4 & 0 \end{bmatrix}$
- Number fields, Quaternions: ???
- Infinity:  $\infty$

# Course Topics: Numbers



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